

ATA BERK ÖZTÜRK

Physics Engineer

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PROFILE

I model the electronic and magnetic properties of two-dimensional materials with density functional theory, and run those calculations as batch workloads on Slurm-scheduled clusters using VASP and Quantum ESPRESSO. For my undergraduate thesis I merged two DFT databases into a 141,171-material dataset and built a classifier that predicts magnetic ordering from chemical composition. I worked as a researcher on a completed TÜBİTAK 2209-A grant on the functionalisation of Janus monolayers.

EDUCATION

Ankara University — B.Sc. in Physics Engineering (English)

2022 – 2026

Faculty of Engineering, Ankara | GPA: 3.13 / 4.00

Ankara University — M.Sc. in Physics Engineering (with thesis)

2026 – present

Graduate School of Natural and Applied Sciences, Ankara | Advisor: Prof. Dr. Yeşim Moğulkoç

EXPERIENCE

Researcher — Computational Condensed Matter Physics Group

Feb 2024 – present

Ankara University, Department of Physics Engineering | hymf.ankara.edu.tr

- Compute band structures, densities of states and magnetic moments of two-dimensional materials with **VASP**; build and visualise structures in **VESTA**. Run surface-adsorption studies as batch jobs on a Slurm-scheduled cluster.
- Automated input generation, job submission and output parsing in **Bash** and **Python**.

High-Performance Computing Intern — EDULINE

Jul – Aug 2025

Istanbul | 20 working days | NVIDIA Elite, Dell EMC Gold and BeeGFS Platinum partner

- Built **Quantum ESPRESSO 7.4.1** from source on Rocky Linux 8.10 (OpenMPI, OpenBLAS, FFTW3, ScaLAPACK, HDF5, libxc) and compared it against an **Intel oneAPI** build. Traced an "Illegal instruction" build failure to its root cause: the virtualised CPU had no **AVX** support.
- Benchmarked core scaling (1/2/4/8 cores) on a representative 2D-materials DFT workload, plotted the scaling curves in **Python/matplotlib** and compared them against published reference figures. Raising VM memory from 8 GB to 32 GB cleared an out-of-memory failure on the 8-core run.
- Configured a single-node **Slurm** / **OpenHPC** environment (slurmctld, slurmd, **munge**, xCAT) and ran scientific software in **Docker** and **Apptainer** containers. Prepared three technical documents covering the installation procedures.

Intern — Institute of Nuclear Sciences

Aug – Sep 2025

Ankara University — Ankara

- Carried out nuclear laboratory experiments, then analysed and reported the resulting measurements. Compiled and configured the **Geant4**, **ROOT** and **Radware** distributions from source on Linux.

PROJECTS

TÜBİTAK 2209-A — National Undergraduate Research Grant

2025 – 2026

Researcher — Ankara University

- "Functionalisation of Two-Dimensional Janus Materials via Density Functional Theory": investigating the surface functionalisation of SnSSe and TiSSe Janus monolayers with DFT.

B.Sc. Thesis — Machine Learning for Magnetic Materials

2025 – 2026

Poster Award, Physics Engineering Thesis Presentations (Jan 2026)

- Merged the C2DB and V2DB databases into a **141,171-material dataset** and classified magnetic ordering with **Gradient Boosting**, reaching **ROC-AUC 0.9863** and **90.6% recall**. Among 2,036 newly generated candidate compounds, the model identified **377 high-confidence magnetic candidates**. Code: github.com/eigen-ml/2d-magnetic-materials-ml

TEKNOFEST International UAV Competition — Team ORION

2022 – 2023

- Contributed as a team member to the application and preliminary evaluation stage of the 2023 competition.

TECHNICAL SKILLS

Computational Physics

VASP, Quantum ESPRESSO, VESTA, Geant4, ROOT, Radware, Qiskit

HPC & Infrastructure

Linux (Rocky, Ubuntu), Slurm, OpenHPC, munge, xCAT, MPI/OpenMPI, OpenMP, Intel oneAPI, OpenBLAS/FFTW3/ScaLAPACK, Apptainer, Docker, Anaconda, Git

Programming

Python (NumPy, Pandas, scikit-learn, matplotlib), C++, Bash, Fortran, MATLAB

Engineering & Reporting

ANSYS (Zemax, OpticStudio, SpaceClaim, Maxwell), AutoCAD, LaTeX, MS Office

CERTIFICATIONS AND LANGUAGES

ARCHER2 HPC Driving Test (University of Edinburgh, 2025) · **Optical Simulation Specialist Training Camp, Distinction** (NUMESYS / ANSYS, 2025) · **QTürkiye Bronze** (Qiskit, 2024) · **TUSAŞ LIFT UP** (2025) · **Physics Summer School** (Bilkent, 2025)

Turkish (native) · English (advanced; B.Sc. taught 100% in English)

REFERENCES

Prof. Dr. Yeşim Moğulkoç (Ankara University) · Prof. Dr. Aybey Moğulkoç (Ankara University) · Zehra Durmuş (Internship Supervisor, EDULINE).